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SRN: PES2UG24CS311
Week: 8
Class: F

1. Register:

a) Test-Bench File:

```
regalutb.v x
regalutb.v
1 `timescale 1 ns / 100 ps
2 `define TESTVECS 8
3
4 module tb;
5     reg clk, reset, wr;
6     reg [2:0] rd_addr_a, rd_addr_b, wr_addr; reg [15:0] d_in;
7     wire [15:0] d_out_a, d_out_b;
8     reg [28:0] test_vecs [0:('TESTVECS-1)];
9     integer i;
10    initial begin $dumpfile("regalutb.vcd"); $dumpvars(0,tb); end
11    initial begin reset = 1'b1; #12.5 reset = 1'b0; end
12    initial clk = 1'b0; always #5 clk =~ clk;
13    initial begin
14        test_vecs[0][28] = 1'b0; test_vecs[0][27] = 1'b1; test_vecs[0][26:25] = 2'bxx; test_vecs[0][24:22] = 3'ox; test_vecs[0][21:19] = 3'ox;
15        test_vecs[1][28] = 1'b0; test_vecs[1][27] = 1'b1; test_vecs[1][26:25] = 2'bxx; test_vecs[1][24:22] = 3'ox; test_vecs[1][21:19] = 3'ox;
16        test_vecs[2][28] = 1'b0; test_vecs[2][27] = 1'b1; test_vecs[2][26:25] = 2'bxx; test_vecs[2][24:22] = 3'o3; test_vecs[2][21:19] = 3'o3;
17        test_vecs[3][28] = 1'b0; test_vecs[3][27] = 1'b1; test_vecs[3][26:25] = 2'bxx; test_vecs[3][24:22] = 3'o1; test_vecs[3][21:19] = 3'o1;
18        test_vecs[4][28] = 1'b0; test_vecs[4][27] = 1'b0; test_vecs[4][26:25] = 2'bxx; test_vecs[4][24:22] = 3'o1; test_vecs[4][21:19] = 3'o1;
19        test_vecs[5][28] = 1'b1; test_vecs[5][27] = 1'b1; test_vecs[5][26:25] = 2'b00; test_vecs[5][24:22] = 3'o1; test_vecs[5][21:19] = 3'o1;
20        test_vecs[6][28] = 1'b1; test_vecs[6][27] = 1'b1; test_vecs[6][26:25] = 2'b01; test_vecs[6][24:22] = 3'o2; test_vecs[6][21:19] = 3'o2;
21        test_vecs[7][28] = 1'b1; test_vecs[7][27] = 1'b0; test_vecs[7][26:25] = 2'b01; test_vecs[7][24:22] = 3'o4; test_vecs[7][21:19] = 3'o4;
22    end
23    initial {wr,rd_addr_a, rd_addr_b, wr_addr, d_in} = 0;
24    reg_file reg_alu_0 (clk, reset, wr, rd_addr_a, rd_addr_b, wr_addr, d_in,d_out_a, d_out_b);
25    initial begin
26        #6 for(i=0;i<'TESTVECS;i=i+1)
27            begin #10 {wr,rd_addr_a, rd_addr_b, wr_addr, d_in}=test_vecs[i];
28        end
29        #100 $finish;
30    end
31 endmodule
32
```

b) Register File:

```
regalutb.v x register.v x
register.v
1 module dftrl_16 (input wire clk, reset, load, input wire [15:0] in, output wire [15:0] out);
2     dftrl_f0(clk, reset, load, in[0], out[0]);
3     dftrl_f1(clk, reset, load, in[1], out[1]);
4     dftrl_f2(clk, reset, load, in[2], out[2]);
5     dftrl_f3(clk, reset, load, in[3], out[3]);
6     dftrl_f4(clk, reset, load, in[4], out[4]);
7     dftrl_f5(clk, reset, load, in[5], out[5]);
8     dftrl_f6(clk, reset, load, in[6], out[6]);
9     dftrl_f7(clk, reset, load, in[7], out[7]);
10    dftrl_f8(clk, reset, load, in[8], out[8]);
11    dftrl_f9(clk, reset, load, in[9], out[9]);
12    dftrl_f10(clk, reset, load, in[10], out[10]);
13    dftrl_f11(clk, reset, load, in[11], out[11]);
14    dftrl_f12(clk, reset, load, in[12], out[12]);
15    dftrl_f13(clk, reset, load, in[13], out[13]);
16    dftrl_f14(clk, reset, load, in[14], out[14]);
17    dftrl_f15(clk, reset, load, in[15], out[15]);
18 endmodule
19
20 module mux8_16 (input wire [0:15] i0, i1, i2, i3, i4, i5, i6, i7, input wire [0:2] j, output wire [0:15] o);
21     mux8_mux8_0(i0[0], i1[0], i2[0], i3[0], i4[0], i5[0], i6[0], i7[0], j[0], j[1], j[2], o[0]);
22     mux8_mux8_1(i0[1], i1[1], i2[1], i3[1], i4[1], i5[1], i6[1], i7[1], j[0], j[1], j[2], o[1]);
23     mux8_mux8_2(i0[2], i1[2], i2[2], i3[2], i4[2], i5[2], i6[2], i7[2], j[0], j[1], j[2], o[2]);
24     mux8_mux8_3(i0[3], i1[3], i2[3], i3[3], i4[3], i5[3], i6[3], i7[3], j[0], j[1], j[2], o[3]);
25     mux8_mux8_4(i0[4], i1[4], i2[4], i3[4], i4[4], i5[4], i6[4], i7[4], j[0], j[1], j[2], o[4]);
26     mux8_mux8_5(i0[5], i1[5], i2[5], i3[5], i4[5], i5[5], i6[5], i7[5], j[0], j[1], j[2], o[5]);
27     mux8_mux8_6(i0[6], i1[6], i2[6], i3[6], i4[6], i5[6], i6[6], i7[6], j[0], j[1], j[2], o[6]);
28     mux8_mux8_7(i0[7], i1[7], i2[7], i3[7], i4[7], i5[7], i6[7], i7[7], j[0], j[1], j[2], o[7]);
29     mux8_mux8_8(i0[8], i1[8], i2[8], i3[8], i4[8], i5[8], i6[8], i7[8], j[0], j[1], j[2], o[8]);
30     mux8_mux8_9(i0[9], i1[9], i2[9], i3[9], i4[9], i5[9], i6[9], i7[9], j[0], j[1], j[2], o[9]);
31     mux8_mux8_10(i0[10], i1[10], i2[10], i3[10], i4[10], i5[10], i6[10], i7[10], j[0], j[1], j[2], o[10]);
32     mux8_mux8_11(i0[11], i1[11], i2[11], i3[11], i4[11], i5[11], i6[11], i7[11], j[0], j[1], j[2], o[11]);
33
```

```
regalutb.v  register.v x
register.v
20 module mux8_16 (input wire [0:15] i0, i1, i2, i3, i4, i5, i6, i7, input wire [0:2] j, output wire [0:15] o);
28     mux8 mux8_7({i0[7], i1[7], i2[7], i3[7], i4[7], i5[7], i6[7], i7[7]}, j[0], j[1], j[2], o[7]);
29     mux8 mux8_8({i0[8], i1[8], i2[8], i3[8], i4[8], i5[8], i6[8], i7[8]}, j[0], j[1], j[2], o[8]);
30     mux8 mux8_9({i0[9], i1[9], i2[9], i3[9], i4[9], i5[9], i6[9], i7[9]}, j[0], j[1], j[2], o[9]);
31     mux8 mux8_10({i0[10], i1[10], i2[10], i3[10], i4[10], i5[10], i6[10], i7[10]}, j[0], j[1], j[2], o[10]);
32     mux8 mux8_11({i0[11], i1[11], i2[11], i3[11], i4[11], i5[11], i6[11], i7[11]}, j[0], j[1], j[2], o[11]);
33     mux8 mux8_12({i0[12], i1[12], i2[12], i3[12], i4[12], i5[12], i6[12], i7[12]}, j[0], j[1], j[2], o[12]);
34     mux8 mux8_13({i0[13], i1[13], i2[13], i3[13], i4[13], i5[13], i6[13], i7[13]}, j[0], j[1], j[2], o[13]);
35     mux8 mux8_14({i0[14], i1[14], i2[14], i3[14], i4[14], i5[14], i6[14], i7[14]}, j[0], j[1], j[2], o[14]);
36     mux8 mux8_15({i0[15], i1[15], i2[15], i3[15], i4[15], i5[15], i6[15], i7[15]}, j[0], j[1], j[2], o[15]);
37 endmodule
38
39 module reg_file (input wire clk, reset, wr, input wire [2:0] rd_addr_a, rd_addr_b, wr_addr, input wire [15:0] d_in, output wire [15:0] d_out_a, d_out_b);
40     wire [7:0] load;
41     wire [15:0] dout_0, dout_1, dout_2, dout_3, dout_4, dout_5, dout_6, dout_7;
42     wire cout;
43     dftrl_16 dftrl_16_0(clk, reset, load[0], d_in, dout_0);
44     dftrl_16 dftrl_16_1(clk, reset, load[1], d_in, dout_1);
45     dftrl_16 dftrl_16_2(clk, reset, load[2], d_in, dout_2);
46     dftrl_16 dftrl_16_3(clk, reset, load[3], d_in, dout_3);
47     dftrl_16 dftrl_16_4(clk, reset, load[4], d_in, dout_4);
48     dftrl_16 dftrl_16_5(clk, reset, load[5], d_in, dout_5);
49     dftrl_16 dftrl_16_6(clk, reset, load[6], d_in, dout_6);
50     dftrl_16 dftrl_16_7(clk, reset, load[7], d_in, dout_7);
51     demux8 demux8_0(wr, wr_addr[2], wr_addr[1], wr_addr[0], load);
52     mux8_16 mux8_16_9(dout_0, dout_1, dout_2, dout_3, dout_4, dout_5, dout_6, dout_7, rd_addr_a, d_out_a);
53     mux8_16 mux8_16_10(dout_0, dout_1, dout_2, dout_3, dout_4, dout_5, dout_6, dout_7, rd_addr_b, d_out_b);
54 endmodule
55
56
```

C)GtkWave:

